Deep Spectral Clustering With Regularized Linear Embedding for Hyperspectral Image Clustering

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***Abstract*— The past decade has witnessed the rapid development of deep learning techniques, especially for large-scale and complex datasets. However, it is still a noteworthy problem in dealing with unsupervised hyperspectral image (HSI) segmentation since inefficiency and misleading result from the absence of supervised information. Generally, spectral clustering (SC) is one of the most powerful clustering algorithms, as it often outperforms other methods for image segmentation. Unfortunately, the poor scalability and generalization severely limit the use of SC, especially for large-scale and high-dimensional HSIs processing. The major motivation of this work is to solve this problem, and we designed a novel algorithm, termed deep SC with regularized linear embedding (DSCRLE), to benefit from both spectral graph theory and deep learning techniques. The brief procedure is first to construct a fully connected neural network to extract latent feature representations, and then normalize the feature representations by the spectral orthonormal constraint. Lastly, by introducing low-dimensional embedding, we refined the final outputs of all given unlabeled hyperspectral pixels. Extensive experiments have demonstrated that the competitiveness of the proposed method, and it outperforms the state-of-the-art clustering approaches in the task of HSI segmentation.**

***Index Terms*— Deep neural network, hyperspectral image (HSI) clustering, spectral graph theory.**

I. Introduction

HYPERSPECTRAL remote sensing techniques have made great achievements along with the development of aerospace and aviation, as it provides hundreds of continuous narrow spectral waves that contain rich spatial, spectral, and temporal information to recognize subtle differences in earth observation [1]. Generally, hyperspectral image (HSI) can be considered as a large-scale 3-D tensor with hundreds of hyperspectral bands, and it needs plenty of labeled pixels for supervised learning to avoid the Hughes phenomenon caused by insufficient training data and underfitting problems. Unfortunately, data annotation is commonly labor-intensive, time-consuming, and extremely expensive, and therefore, it has become one of the bottlenecks that limits the wide applications of HSI processing [2]. Alternatively, HSI clustering is a potential technique for HSI intelligent interpretation without manually labeled samples and expensive prior knowledge.

Traditional clustering algorithms, including k-means (KM), spectral clustering (SC), and Gaussian mixture model (GMM), have been successfully used in many fields of artificial intelligent including machine learning and image processing [3]. Recently, graph-based clustering algorithms, including SC [4], are leading and prevalent methods in academia and industry because they provide reasonable mathematical models referring to manifold learning. Since SC is solved by an eigen decomposition of graph Laplacian matrix, large-scale HSI direct computation of eigenvectors may be prohibitive. To solve this challenging problem, Nyström extension [5] based on local similarities and low-rank matrix approximation has been widely introduced by constructing an approximated affinity matrix. Fast SC with anchor graph (FSCAG) [6] proposed an efficient solution to reconstruct the regression matrix by bipartite graph for HSI segmentation. Projected clustering with adaptive neighbors (PCAN)-assigned [7] adaptive neighbors refer to anchor-based graph to obtain a robust affinity matrix. Multi-task spectral clustering based on iterative optimization (MSCIO) [8] solved the eigen decomposition problem of multitask SC by introducing an iterative optimization. However, SC-based algorithms have a bottleneck that they cannot provide a natural extension to cope with unseen data points. To solve this problem, Alzate and Suykens [9] introduced an additional constraint to correct features for the out-of-sample extension. Li et al. [10] proposed a subspace clustering approach based on the Cauchy loss function to get rid of the influences of noisy data. Spectral embedded clustering (SEC) [11] added a linearity regularization into the vanilla objective function of SC, and it can naturally deal with unseen data points.

However, the hand-crafted features designed by traditional clustering algorithms may not be used for synthesizing many different image classification tasks. The past decade has witnessed the rapid development of deep learning techniques, especially in solving problems concerning large-scale and complex data. However, vanilla deep learning models require accurately labeled samples for training, and it is still a tough problem to cope with HSIs, since the absence of supervised information. Accordingly, deep learning-based (DL-based) unsupervised learning techniques

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for HSI processing have recently been developed considerably. The existing approaches can be summarized into three main branches including autoencoder-based clustering (AEC) methods, separated network-based clustering methods, and Siamese network-based clustering methods according to [12].

1) AEC [13] optimized a mapping function according to the reconstruction error between output code and input data. Deep embedded clustering (DEC) [14] iteratively optimized a Kullback-Leibler (KL) divergence-based objective function with a stacked autoencoder network to obtain latent representations. Variational autoencoder (VAE) [15] endeavored to model a generative process by latent representation learning to approximate the posterior distribution. Variational deep embedding (VaDE) [16] solved the clustering problems by combining deep neural networks and GMMs. VAE with distance (VAED) [17] further extended AEC and VAE to solve the vanishing gradient problem of deep learning. Li et al. [18] embedded a structured graph with adaptive neighbors into deep autoencoder networks. Tulczyjew et al. [19] proposed recurrent neural network-based asymmetric autoencoders to learn the compressed representation of unlabeled data. However, the above AEC algorithms have not been satisfactory, because their reconstruction loss could ignore the discriminative ability of latent representations.
2) Separated network-based clustering methods are composed of deep neural networks and unsupervised learning constraints. Deep spectral clustering learning (DSCL) [20] proposed a metric learning framework to obtain the latent representations of unlabeled data. SC via ensemble deep autoencoder learning (SC-EDAE) [21] successfully applied SC on an ensemble of fused encodings obtained from deep autoencoder networks. However, it simply combined SC and deep autoencoder networks by a reconstruction loss function, and this needs to be more satisfactory. SpectralNet (SpecNet) [22] learned a mapping function to embed data points into the eigenspace of Laplacian matrix with a fully connected neural network. Nonetheless, it is still facing the same problem with SC, the objective function of which relies on the manifold assumption, and high-dimensional or sparse datasets may break this assumption.
3) Siamese network-based clustering methods take in similar input, produce comparable output, and the model can be optimized by contrastive loss function [23]. Motivated by contrastive clustering (CC) [24], Cai et al. [25] proposed spectral–spatial CC (SSCC) for HSI segmentation, and it is illustrated that data augmentation operations show great potential for clustering performance. It may be noted that this type of clustering approach is highly dependent on the performance of data argumentation, and therefore, they are sensitive to noise samples.

Motivated by the above methods, we designed an efficient and effective general framework, named deep SC with regularized linear embedding (DSCRLE), and take advantage of spectral graph theory and deep learning techniques for unsupervised learning. Our primary strategy is first to utilize a fully connected neural network to extract the latent feature representations of HSIs, and then normalize the latent feature representations by spectral orthonormal constraints. Lastly, we further refine the latent feature representations by introducing regularized linear embedding. The quantitative evaluation reported that the performance of our proposed DSCRLE is significantly superior to the state-of-the-art methods in dealing with large-scale HSIs.

The novelty and contribution of DSCRLE could be summarized as below.

1) SC-based algorithms greatly depend on learning manifold relationships and local similarities for robust affinity matrix construction, but high-dimensional or sparse data distribution may break this assumption. Due to this limitation, we explicitly introduced regularized linear embedding for solving the disruption of high-dimensional and sparse data distribution, and the proposed objective function took into account both manifold structure information and regularized linear embedding constraints.
2) Another problem of SC-based algorithms is limited scalability, which is the major limitation of applying them in real-world datasets. In this work, we implemented a fully connected neural network with spectral orthogonality constraints, and it could perform the optimization in a stochastic fashion. It obtained a feasible solution for the eigenvalue decomposition of SC, and this operation could definitely be available for scalability and generalization.
3) Out-of-sample-extension (OOSE) is another tough problem, as SC cannot provide a natural extension for unseen data points. By comparison, DSCRLE is a general deep learning framework, and once training has finished, it can naturally embed new test data points without any extra training. We compared DSCRLE with some popular clustering algorithms, including DL-based and nondeep learning based on real-world HSI datasets, and extensive experiments have demonstrated that the proposed algorithm outperforms others.

II. Related Works

A. Preliminaries

In this part, we briefly introduce some fundamental theories of our motivation, including SC, regularized linear classifier, and deep clustering in this section. Before that, we present some significant notations to ensure that their mathematical meanings have been formulated clearly. The upper case bold letters denote matrices such as identity matrix $\mathbf{I}$ and affinity/similarity matrix $\mathbf{S}$. The lower case bold letters denote vectors such as cluster assignment vector $\mathbf{y}_i$, and the lower case mathematical letters denote scalars. $||\cdot||_2$ means the L2 norm and trace$(\cdot)$ demonstrates the trace operation.

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B. Anchor-Based Graph

The vanilla SC adopts an affinity/similarity matrix $\mathbf{S}$ via a simple Gaussian distance function as below

$$s_{ij} = \begin{cases} e^{\frac{-||\mathbf{x}_i - \mathbf{x}_j||_2^2}{2\sigma^2}}, & \text{if } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where s_{ij} is the similarity of sample $\mathbf{x}_i$ and sample $\mathbf{x}_i$, and σ controls the scale of neighbors. Note that, $\mathbf{x}_i$ and $\mathbf{x}_j$ are neighbors i.i.f. $\mathbf{x}_i$ is among k nearest neighbors of $\mathbf{x}_j$ and vice versa.

It is worth noting that the performance of vanilla SC are heavily affected due to the sensitivity of parameter σ, and some recent studies improve the discrimination of affinity matrix by anchor-based graph [26]. Denote $\mathbf{Z} \in \mathbb{R}^{n\times m}$ as the cross and sparse similarity matrix between anchors sets $\mathbf{X}$ and data points set $\mathbf{A}$. We can define $h(\mathbf{x}_i, \mathbf{a}_i) = ||\mathbf{x}_i - \mathbf{a}_j||_2^2$, which is the square of Euclidean distance between data point $\mathbf{x}_i$ and anchor $\mathbf{a}_i$. z_{ij} can be represented as

$$z_{ij} = \frac{\hat{h}(\mathbf{x}_i, \mathbf{a}_{K+1}) - \hat{h}(\mathbf{x}_i, \mathbf{a}_j)}{K\hat{h}(\mathbf{x}_i, \mathbf{a}_{K+1}) - \sum_{k=1}^{K} \hat{h}(\mathbf{x}_i, \mathbf{a}_k)} \tag{2}$$

where $\hat{h}_i = \theta(h_i)$ which sorts the distance in ascending order.

C. Regularized Linear Classifier

SC is good, but this type of spectral graph theory-based algorithms may not perform well when dealing with high-dimensional or sparse data points as the manifold information may not be well-defined [27]. According to [28], the cluster assignment matrix can be defined by a low-dimensional mapping function with least square. Specifically, for the two-class problem, it expects to learn a column vector $\mathbf{w}$ such that the squared cost

$$\mathcal{J} = \min \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i^T \mathbf{w} - y_i)^2 + \mu ||\mathbf{w}||^2 \tag{3}$$

where μ is a regularization parameter and $y_i \in \{+1, -1\}$. It is noteworthy that $||\mathbf{w}||^2$ is a regularization term for avoiding the singularity of $\mathbf{X}\mathbf{X}^T$ (e.g., when the dimension of data is larger than the number of data).

For multiclass problem, we can introduce one classifier to each class in a similar way, and (3) can be represented as

$$\mathcal{J} = \min \frac{1}{n} \sum_{i=1}^{n} ||\mathbf{x}_i^T \mathbf{w}^c - \mathbf{y}_i||_2^2 + \mu ||\mathbf{w}^c||^2 \tag{4}$$

where $\mathbf{y}_i \in \mathbb{R}^C$ is a one-hot vector, and C is the number of clusters. If $\mathbf{x}_i$ belongs to class, the cth element of $\mathbf{y}_i$ is 1, and others are 0. By setting the derivatives of (4) with respect to $\mathbf{w}^c$ (e.g., $\partial\mathcal{J}/\partial\mathbf{w}^c = 0$), we can get the optimal solution of (4) as

$$(\mathbf{w}^c)^* = (\mathbf{X}\mathbf{X}^T + \mu n \mathbf{I})^{-1} \mathbf{X}\mathbf{y}^c \tag{5}$$

where $\mathbf{I}$ is an identity matrix. After learning the linear projections $\mathbf{W}^*$ which is combined with $(\mathbf{w}^c)^*$, we can finally get the cluster assignment matrix which is close to a linear space spanned by the data $\mathbf{X}$ as

$$\mathbf{Y} = \mathbf{X}^T(\mathbf{W}^*). \tag{6}$$

III. Deep SC With Regularized Linear Embedding

In this section, we elaborate on the specifics of our proposed algorithm, which is termed DSCRLE. First, the proposed method directly incorporates linearity regularization into the SC methods to solve the high-dimensional and small sample-size problems. And then, we construct a fully connected neural network satisfied with orthogonality constraints, and it performs the optimization in a stochastic fashion. Once the training is completed, we can naturally embed new test data points without extra training.

A. Formulation

As illustrated in Fig. 1, the framework of DSCRLE can be divided into two parts. In the first part, we randomly sample a batch of data points and learn normalized graph Laplacian matrix $\mathbf{L}$ which is referred to anchor graph construction (AGC). The second part implements a feed-forward network that contains both manifold structure information and regularized linear embedding constraints.

The objective function of DSCRLE can be denoted as

$$\mathcal{L} = \mathcal{L}_{\text{SC}} + \lambda \mathcal{L}_{\text{RLC}} \tag{7}$$

where $\mathcal{L}_{\text{SC}}$ is the objective function of SC with normalized graph Laplacian, $\mathcal{L}_{\text{RLC}}$ is the objective function of regularized linear embedding, and λ is the tradeoff parameter to balance their weights.

1) Anchor Graph Construction: According to [26], affinity matrix $\mathbf{S}$ can be considered as a variable and solved by

$$\min_{\mathbf{s}_i^T \mathbf{1}=1, s_{ij}\geq 0} \sum_{i,j=1}^{n} s_{ij} ||\mathbf{x}_i - \mathbf{x}_j||_2^2 + \gamma s_{ij}^2 \tag{8}$$

where γ is the regularization parameter. Considering the affinity matrix $\mathbf{S}$ is defined to estimate the similarity among samples, s_{ij} should be larger than zero. Denote $d_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||_2^2$, and (8) can be rewritten in vector form as

$$\min \frac{1}{2} \left\| \mathbf{s}_i + \frac{\mathbf{d}_i}{2\gamma} \right\|_2^2, \quad \text{s.t. } \mathbf{s}_i^T \mathbf{1} = 1, \ s_{ij} \geq 0. \tag{9}$$

For each $\mathbf{s}_i$, (9) can be solved by introducing Lagrangian multipliers, and the parameter γ can be set as

$$\gamma = \frac{k}{2} d_{i,k+1} - \frac{1}{2} \sum_{j=1}^{k} d_{ij}. \tag{10}$$

Substituting (10) into (9), and the solution of (8) can be presented as

$$s_{ij} = \frac{d_{i,K+1} - d_{ij}}{K d_{i,K+1} - \sum_{k=1}^{k} d_{ik}}. \tag{11}$$

Different from the vanilla affinity matrix of SC, the affinity matrix $\mathbf{S}$ with AGC is scale invariant, and it is naturally K-sparse since only the top K samples are considered, and the others are set to zero.

$\mathcal{L}_{\text{SC}}$ is the criterion for AGC, which encapsulates local invariance on manifold and representation consistency as follows:

$$\mathcal{L}_{\text{SC}} = \frac{1}{n^2} \sum_{ij}^{n} s_{ij} ||\mathbf{y}_i - \mathbf{y}_j||_2^2 \tag{12}$$

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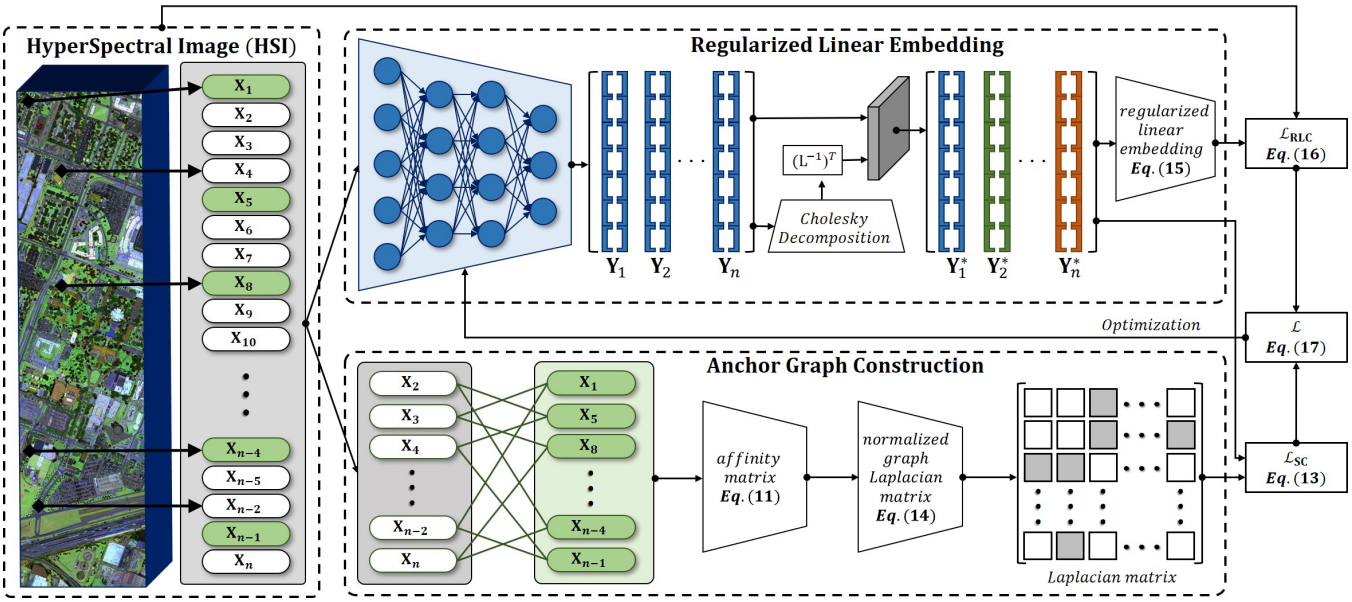

Fig. 1. Flowchart of the proposed method.

where s_{ij} is the elements of affinity matrix $\mathbf{S}$ referred in (11).

It is obvious that (12) is equivalent to the following equation according to spectral graph theory:

$$\mathcal{L}_{\mathrm{SC}} = \mathrm{trace}\,(\mathbf{Y}^T\mathbf{L}\mathbf{Y}) \tag{13}$$

where $\mathbf{Y}^T\mathbf{Y} = \mathbf{I}$ and $\mathbf{L}$ is a symmetric matrix referred to normalized anchor Laplacian matrix represented as

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{S}\mathbf{D}^{-\frac{1}{2}}. \tag{14}$$

Different from [26], the above affinity matrix calculates the similarities from data points in batches. Therefore, the batch size of each iteration must be large enough, as the affinity matrix $\mathbf{S}$ needs to capture the manifold information of the global dataset $\mathbf{X}$. The larger the batch size, the more time and space we need to take for affinity matrix construction. DSCRLE, therefore, trades accuracy with scalability and generalization ability.

2) Regularized Linear Embedding: Vanilla SC partitions' data points highly depend on the manifold information of data, but in some high-dimensional and sparse datasets, the manifold information may need to be better defined. A good choice is to take into account both manifold structure information and regularized linear embedding. According to [28], the cluster assignment matrix can be defined by a low-dimensional mapping function, especially for solving the high-dimensional and small sample-size problems. $\mathcal{L}_{\mathrm{RLC}}$ is the criterion for regularized linear embedding which can be denoted as

$$\mathcal{L}_{\mathrm{RLC}} = ||\mathbf{X}^T\mathbf{W} - \mathbf{Y}||^2 + \mu||\mathbf{W}||^2 \tag{15}$$

where $\mathbf{W}^* = (\mathbf{X}\mathbf{X}^T + \mu\mathbf{I})^{-1}\mathbf{X}\mathbf{Y}$ according to [28]. Substituting $\mathbf{W}^*$ into (15) and we can get

$$\mathcal{L}_{\mathrm{RLC}} = \mathrm{trace}\,(\mathbf{Y}^T\mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \mu\mathbf{I})^{-1}\mathbf{X}\mathbf{Y}). \tag{16}$$

For this perspective, we can substitute $\mathcal{L}_{\mathrm{SC}}$ and $\mathcal{L}_{\mathrm{RLC}}$ into (7) and $\mathcal{L}$ can be transformed as

$$\begin{aligned}\mathcal{L} &= \mathrm{trace}\,(\mathbf{Y}^T\mathbf{L}\mathbf{Y}) + \lambda(\mathrm{trace}\,(\mathbf{Y}^T\mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \mu\mathbf{I})^{-1}\mathbf{X}\mathbf{Y})) \\ &= \mathrm{trace}\,(\mathbf{Y}^T(\mathbf{L} + \lambda(\mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \mu\mathbf{I})^{-1})\mathbf{X})\mathbf{Y})\end{aligned} \tag{17}$$

where $\mathbf{Y}^T\mathbf{Y} = \mathbf{I}$. Obviously, the optimal solution can be obtained by a sample decomposition for $\mathbf{L} + \lambda(\mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \mu\mathbf{I})^{-1})\mathbf{X}$, and it contains both manifold structure information and linear embedding constraints. Similar to SC-based algorithms, the optimal solution can be learned by a simple spectral decomposition and the final discrete solution for cluster assignment is obtained by KM.

However, there are two issues deserving particular attention. First, limited by the space and time complexity of affinity matrix construction, SC-based algorithms are not available for large-scale datasets. The space complexity is $O(N^2)$ and the time complexity is $O(N^2d)$, and this is the main obstacle to real-world datasets. Second, SC-based algorithms do not provide a natural extension for unseen data points. To overcome this shortcoming, we replace the procedure of spectral decomposition with a fully connected neural network.

B. Optimization

In general, the solution of (17) can be relaxed as the eigenvectors of $\mathbf{L}+\lambda(\mathbf{X}^T(\mathbf{X}\mathbf{X}^T+\mu\mathbf{I})^{-1})\mathbf{X}$. We expect to learn a fully connected neural network to approximate the eigenvalue decomposition of SC, which could be easily optimized by the back propagation. Therefore, we construct a fully connected neural network with only five fully connected layers to learn the label matrix $\mathbf{Y}$. Our experiments demonstrate that a five-layer fully connected neural network is good enough for HSI datasets. But there is a problem with the output of vanilla deep neural network as it has to satisfy the orthonormal constraint of SC as $\mathbf{Y}^T\mathbf{Y} = \mathbf{I}$.

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To solve this problem, DSCRLE is trained with a spectral orthogonality constraint. This can be easily implemented by adding a linear transform layer, whose weights are assigned by Cholesky decomposition. We introduce a linear transform function at the top of the neural network to make sure the output can satisfy the orthonormal constraint, and this can be defined as

$$\mathbf{Y}^{\star} = \mathbf{\Gamma}(\mathbf{Y}) \tag{18}$$

where $\mathbf{\Gamma}$ is the linear transform function and $\mathbf{Y}^{\star}$ has the same shape as $\mathbf{Y}$.

Motivated by [22], the linear transform function $\mathbf{\Gamma}(\cdot)$ can be optimized through the following theorem.

Proposition 1: Given a matrix $\mathbf{A}$. The Cholesky decomposition of matrix $\mathbf{A}^T\mathbf{A}$ can be defined as $\mathbf{L}\mathbf{L}^T = \mathbf{A}^T\mathbf{A}$, and $\mathbf{L}$ is a lower triangular matrix. Suppose $\mathbf{A}^T\mathbf{A}$ is a symmetric positive-definite matrix, and we have $\mathbf{Q} = \mathbf{A}(\mathbf{L}^{-1})^T$.

Proof: The QR decomposition of $\mathbf{A}$ is $\mathbf{A} = \mathbf{Q}\mathbf{R}$ where $\mathbf{R}$ is a upper triangular matrix and $\mathbf{Q}$ is a orthogonal matrix. According to Proposition 1, $\mathbf{A}^T\mathbf{A}$ is a symmetric positive-definite matrix, we can get $\mathbf{L}\mathbf{L}^T = \mathbf{A}^T\mathbf{A}$. Note that, if $\mathbf{L}$ is a lower triangular matrix, $\mathbf{L}^T$ would be a upper triangular matrix. It is obvious that $\mathbf{A} = \mathbf{A}(\mathbf{L}^{-1})^T\mathbf{L}^T$, the column space of the first i columns of $\mathbf{A}$ is the same as the columns space of the first i columns of $\mathbf{A}(\mathbf{L}^{-1})^T$. Therefore, if $\mathbf{A}(\mathbf{L}^{-1})^T$ is an orthogonal matrix, the matrix $\mathbf{Q}$ can be denoted as $\mathbf{Q} = \mathbf{A}(\mathbf{L}^{-1})^T$ and the matrix $\mathbf{R}$ is happen to be $\mathbf{L}^T$.

Mathematically, we can get an orthogonal cluster assignment matrix $\mathbf{F}^{\star}$ by multiplying $\sqrt{m}(\mathbf{L}^{-1})^T$ from right as

$$\mathbf{Y}^{\star} = \mathbf{\Gamma}(\mathbf{Y}) = \mathbf{Y}(\sqrt{m}\mathbf{L}^{-1})^T. \tag{19}$$

Obviously, $\mathbf{Y}$ is obtained by Cholesky decomposition where $\mathbf{L}\mathbf{L}^T = \mathbf{Y}^T\mathbf{Y}$, $\sqrt{m}$ is a factor for satisfying $\mathbf{Y}^T\mathbf{Y} = \mathbf{I}$ and $\mathbf{L}$ is a lower triangular matrix.

Algorithm 1 Algorithm to Solve the Problem in Eq. (17)

Input: Dataset $\mathbf{X} \in \mathbb{R}^{d\times n}$, the number of clusters c.
Output: Indicator matrices $\mathbf{Y} \in \mathbb{R}^{n\times c}$.
Initialize a five-layer fully connected neural network $G(\theta)$;
while *Eq. (17) not converge* **do**
1) Construct a batch from $\mathbf{X}$ and calculate the normalized graph Laplacian matrix $\mathbf{L}$ according to Eq. (11) and Eq. (14);
2) Forward propagate the each batch through $G(\theta)$ and obtain the orthogonal output $\mathbf{Y}^{\star}$ according to Eq. (19);
3) Compute $\mathcal{L}_{SC}$ according to Eq. (13);
4) Compute $\mathcal{L}_{RLC}$ according to Eq. (16);
5) Compute $\mathcal{L}$ according to Eq. (17);
6) Use the gradient of $\mathcal{L}$ to update all weights θ, except the linear transform layer at top of network;
end
Once training finished, forward propagate data points to get the orthogonal output, and then obtain final cluster assignment matrix through k-means.

IV. Experiments

This section compares the state-of-the-art clustering algorithms with the proposed DSCRLE on several benchmark HSI datasets.

A. Experiment Setting

Our experiments were implemented in the academic and industry standard numerical computing environment PYTHON (version 3.6.10) on a Linux-based operating system of Ubuntu (version 16.04). The computational environment was set up on a computer with Inter[1] Core[2] i7-6800K 3.40 GHz CPU and GTX 1080 Ti 10G GPU. We carry out our empirical experiments on the following datasets as presented in Table VIII.

1) *MNIST [29][3]:* It contains 70 000 grayscales images of handwritten digits (0–9) obtained from NIST, and the images are 28 × 28 grayscale pixels. It is loaded from the Keras package and we reshaped each image to a 784-dimensional vector.
2) *Indian Pines[4]:* It is a HSI segmentation dataset and it consists of hyperspectral bands over a single landscape in Indiana, U.S., with 145 × 145 pixels and 220 spectral channels. Its available ground-truth data contain 10 366 pixels with 16 classes.
3) *DFTC2013:* An HSI with 1905 × 349 and it consists of 144 spectral bands in the 380 to 1050 nm region, which has been calibrated to at-sensor spectral radiance units. Its spatial resolution is 2.5 m and land cover and land use classes were considered for ground-truth data.
4) *Salinas[5]:* It is collected by the 224-band AVIRIS sensor over Salinas Valley, California, and is characterized by high spatial resolution (3.7-m pixels). The area covered comprises 512 lines by 217 samples. About 20 water absorption bands were discarded: [108-112], [154-167], 224. It includes vegetables, bare soils, and vineyard fields and the ground-truth data contains 16 classes.
5) *Botswana[6]:* Botswana is a HSI dataset consisting of observations from 14 identified classes. Uncalibrated and noisy bands that cover water absorption features were removed, and the remaining 145 bands were included as candidate features.
6) *Pavia Centre[7]/University[8]:* Pavia Centre/University are two benchmark hyperspectral datasets acquired by the ROSIS sensor during a flight campaign over Pavia, northern Italy. The number of spectral bands is 102 for

[1]Registered trademark.
[2]Trademarked.
[3]https://deepai.org/dataset/mnist
[4]https://paperswithcode.com/dataset/indian-pines
[5]https://paperswithcode.com/dataset/salinas
[6]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes
[7]https://paperswithcode.com/dataset/pavia-centre
[8]https://paperswithcode.com/dataset/pavia-university

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Pavia Centre and 103 for Pavia University. Pavia Centre is a 1096×1096 pixels image, and Pavia University is 610×610 pixels. The geometric resolution is 1.3 m, and both image ground truths differentiate 9 classes each.

B. Evaluation Metrics

We adopt five common measures to evaluate the tested algorithms for a comprehensive investigation. Accuracy is computed for the best matching permutation between ground-truth and predicted labels. Purity measures the extent to which a group of records shares the same class. Normalized mutual information (NMI) measures the agreement of the two assignments, ignoring permutations. Kappa is a statistic that measures inter-annotator agreement. V-measure evaluates the agreement of two independent assignments on the same dataset.

For all metrics, the higher the value, the better the performance.

1) ACCuracy (ACC.) is defined as
$$\text{ACC.} = \max_m \frac{\sum_{i=1}^{n} \mathbf{1}\{l_i = m(c_i)\}}{n} \tag{20}$$
where c_i is the estimated label, l_i is the true label, and τ ranges over all permutations of cluster labels.
2) Purity (P.) is defined as
$$\text{P.} = \sum_{i=1}^{K} \frac{n_i}{n} P(c_i) \tag{21}$$
where $P(c_i)$ is the purity of a single cluster, defined as $P(c_i) = \max_j p_{ij}$.
3) NMI quantifies how much the predicted labels are informative about the true labels as
$$\text{NMI} = \frac{\sum_{i=1}^{c}\sum_{j=1}^{c} n_{i,j} \log \frac{n_{i,j}}{n_i \hat{n}_j}}{\sqrt{\left(\sum_{i=1}^{c} n_i \log \frac{n_i}{n}\right)\left(\sum_{i=1}^{c} \hat{n}_i \log \frac{\hat{n}_i}{n}\right)}} \tag{22}$$
where $\hat{n}_j$ is the number of data volumes belonging to the group $\mathcal{L}_j (1 \le j \le c)$, n_i is the number of data volumes contained in the group $\mathcal{C}_i (1 \le i \le c)$, and $n_{i,j}$ denotes the number of data volumes which are in the intersection of both the group $\mathcal{L}_i$ and the group $\mathcal{C}_j$.
4) Kappa (K.) is defined as
$$\text{K.} = \frac{p_o - p_e}{1 - p_e} \tag{23}$$
where p_o is the empirical probability of agreement on the label assigned to any sample (the observed agreement ratio), and p_e is the expected agreement when both annotators assign labels randomly.
5) V-measure (V.) is defined as the harmonic mean of homogeneity h and completeness c of the clustering
$$\text{V.} = (1 + \beta)\frac{h \cdot c}{\beta \cdot h + c} \tag{24}$$
where homogeneity h is maximized when each cluster contains elements of few different classes as possible. Completeness c aims to put all elements of each class in single clusters.

C. Implement Details

All experiments are under the same hardware environment: NVIDIA GTX 1080 Ti with 10G, Python 3.7.0, Pytorch 1.11.0, and Tensorflow 2.6.0. For a fair comparison, the data argumentation is substantially consistent, and we only adapt StandardScaler (provided by Python Scikit-learn ToolKit) without additional dimensionality reduction for data normalization. Meanwhile, the spatial information of HSIs is incorporated to help with the spectral analysis by considering neighboring pixels, to accurately model the spatial contextual information of HSIs. For MNIST dataset, the input images are transformed as 1-D vectors and we also adapt StandardScaler to them. It should be noted that the SSC provided by the Vision Lab of Johns Hopkins University introduces additional data argumentation by adapting wavelet scattering transform normalization. We remove wavelet-scattering transform normalization for a fair comparison, and it is also difficult to be adapted to HSI datasets. However, we believe that wavelet-scattering transform normalization is a potential technique that may be useful for improving the data argumentation of HSI datasets.

For nDL-based algorithms, KM, GMM, and SC are three benchmark clustering methods, following by the default configuration referred to Python Scikit-learn ToolKit. In the experiments, we adapt mini-batch KM as it drastically reduces the computation required to converge to a local solution. The size of the mini-batches in KM is 1024. GMM allows us to estimate the parameters of a Gaussian mixture distribution and we use KM to initialize weights, means, and precisions. SC performs a low-dimensional embedding of the affinity matrix between data points. It constructs the affinity matrix by computing a graph of nearest neighbors, where the number of neighbors is set to 10 as a default. SSC-based algorithms, including EnSC, sparse subspace clustering by orthogonal matching pursuit (SSC-OMP), and SSC with basis pursuit (SSC-BP) follow the default configuration; the parameter τ, which is for elastic net penalty term, is set to 1.0 for SSC-BP and 0.1 for EnSC.

For DL-based algorithms, we set the number of epochs is 100, and the size of each batch is 1024. The optimizer of AEC is Adam and the initial learning rate is set to 0.001. The optimizer of VAE is also Adam, the initial learning rate is set to 0.005, and the weight of KL loss is set to 0.00025. The optimizer of DEC is SGD, the initial learning rate is set to 0.01, and the momentum is set to 0.9. The number of epochs for pretraining is set to 50 and the update interval is set to 10. However, the above parameter settings are not available for MNIST dataset and we follow the default parameter settings, where the number of epochs for pretraining is set to 300, the update interval is set to 140, and the number of epochs is 2000. Another difference is that we use MinMaxScaler here for data argumentation in the experiment of MNIST dataset. The optimizer of Siamese network (SiamNet) is RMSprop and the initial learning rate is set to 0.0001. It may be noted that SiamNet needs to generate positive and negative pairs from unlabeled data. Specifically, it draws from the k nearest neighbors of each point to form pairs, where $k = 2$ in our experiments. The optimizer of SpecNet is RMSprop, the initial

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TABLE I
QUANTITATIVE EVALUATION ON THE MNIST DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.4472	0.4418	0.6835	TO	TO	TO	0.6467	0.2902	0.8940	0.7140	0.9592	**0.9593**
P.	0.5377	0.4686	0.7440	TO	TO	TO	0.6759	0.3258	0.8940	0.7437	0.9562	**0.9593**
RS.	0.8649	0.8421	0.9241	TO	TO	TO	0.8972	0.8419	0.9680	0.9332	0.9842	**0.9843**
NMI	0.4526	0.3750	0.7525	TO	TO	TO	0.6283	0.1995	0.8472	0.7482	**0.8996**	**0.8996**
K.	0.3863	0.3797	0.6483	TO	TO	TO	0.6064	0.2115	0.8822	0.6822	0.9545	**0.9548**
V.	0.4526	0.3750	0.7525	TO	TO	TO	0.6283	0.1995	0.8472	0.7482	**0.8996**	**0.8996**
mean	0.5236	0.4804	0.7508	TO	TO	TO	0.6805	0.3447	0.8888	0.7616	0.9427	**0.9428**

TABLE II
QUANTITATIVE EVALUATION ON THE INDIAN PINES DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.4016	0.3517	0.4633	0.4495	0.3475	**0.4841**	0.3258	0.2226	0.4130	0.3835	0.5100	0.4836
P.	0.5221	0.5310	0.5490	0.5881	0.4012	**0.6530**	0.5122	0.4063	0.5274	0.5591	0.5575	0.5952
RS.	0.8493	0.8544	0.7900	0.8612	0.7989	**0.8752**	0.8565	0.8457	0.8519	0.8671	0.8434	0.8452
NMI	0.4480	0.4385	0.5390	0.5017	0.2827	**0.5572**	0.4102	0.2616	0.4563	0.4617	0.5104	0.5083
K.	0.3691	0.3032	0.4366	0.4106	0.3074	0.4496	0.2775	0.1759	0.3862	0.3409	0.4721	**0.4623**
V.	0.4480	0.4385	0.5390	0.5017	0.2827	**0.5572**	0.4102	0.2616	0.4563	0.4617	0.5104	0.5083
mean	0.5063	0.4862	0.5528	0.5521	0.4034	**0.5961**	0.4654	0.3623	0.5152	0.5123	0.5673	0.5671

TABLE III
QUANTITATIVE EVALUATION ON THE SALINAS DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.7193	0.7461	**0.7977**	0.5618	0.3059	0.6893	0.7092	0.2088	0.7243	0.6327	0.7214	0.7609
P.	0.7646	0.8010	**0.8037**	0.6418	0.3073	0.7050	0.7442	0.4042	0.7617	0.7784	0.7528	0.7962
RS.	0.9269	**0.9355**	0.9322	0.8931	0.4829	0.9141	0.9240	0.8669	0.9283	0.9225	0.9263	0.9202
NMI	0.7708	0.8079	**0.8664**	0.6959	0.1562	0.7805	0.7757	0.2615	0.7868	0.7538	0.8235	0.8504
K.	0.7032	0.7274	**0.7842**	0.5376	0.2670	0.6676	0.6929	0.1580	0.7070	0.6077	0.7059	0.7511
V.	0.7708	0.8079	**0.8664**	0.6959	0.1562	0.7805	0.7757	0.2615	0.7868	0.7538	0.8235	0.8504
mean	0.7759	0.8043	**0.8418**	0.6710	0.2793	0.7562	0.7703	0.3601	0.7825	0.7415	0.7922	0.8215

TABLE IV
QUANTITATIVE EVALUATION ON THE DFTC2013 DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.5862	0.5953	0.5992	0.4742	0.1232	0.4149	0.5297	0.3347	0.6031	0.6684	0.6186	**0.6748**
P.	0.5929	0.6268	0.6529	0.5021	0.1306	0.4371	0.5530	0.3475	0.6299	0.6783	0.6292	**0.6900**
RS.	0.9191	0.9225	0.9177	0.9144	0.3189	0.9009	0.9073	0.8856	0.9257	**0.9422**	0.9285	**0.9422**
NMI	0.6049	0.6665	0.6843	0.5054	0.0959	0.4700	0.6026	0.3529	0.6782	0.7060	0.6859	**0.7096**
K.	0.5595	0.5701	0.5750	0.4389	0.1138	0.3763	0.5009	0.2934	0.5764	0.6468	0.5933	**0.6525**
V.	0.6049	0.6665	0.6843	0.5054	0.0959	0.4700	0.6026	0.3529	0.6782	0.7060	0.6859	**0.7096**
mean	0.6446	0.6746	0.6856	0.5568	0.1464	0.5115	0.6160	0.4278	0.6819	0.7246	0.6902	**0.7298**

learning rate is set to 0.0001, and the number of neighbors for affinity matrix construction is set to 2. Different from the official implementation of SpecNet, we do not use the additional Siamese networks for pretraining and this difference in quality can be quite small in practice. The optimizer of DSCRLE is also RMSprop, the initial learning rate is set to 0.0001, and the top k samples are considered for AGC. Following the experiment settings in [11], the tradeoff parameters λ and μ are set to 1.0 and 1e-5.

D. Comparisons With State of the Arts

The experimental results are illustrated in Tables I–VII which are DL-based and non-DL-based (nDL-based) clustering algorithms. KM, GMM, and SC are three benchmark clustering methods, and we follow the default configuration refer to Python Scikit-learn ToolKit.[9] Elastic net subspace clustering (EnSC) [30], SSC-OMP, [31] and SSC-BP [32] are three vanilla subspace clustering algorithms which are provided by the Vision Lab of Johns Hopkins University.[10] SSC-BP computes sparse representation via solving the basis pursuit, SSC-OMP uses the orthogonal matching pursuit algorithm for computing sparse representations, and EnSC uses a mixture of the L1 regularization (which is used in SSC-BP) and the L2 regularizations for subspace clustering. The remains are all based on DL-based algorithm and their full names and references are AEC [13], VAE [15], [33], DEC [14], SiamNet [34], and SpecNet [22]. AEC is based on autoencoder network which considers data reconstruction and compactness simultaneously. VAE introduces a stochastic variational inference and learning algorithm that scales to large datasets. SiamNet is composed of two networks and is trained to minimize the contrastive loss, which encourages the distance between positive pairs to be small and between

[9]https://scikit-learn.org/stable/modules/clustering.html

[10]https://github.com/ChongYou/subspace-clustering

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TABLE V
QUANTITATIVE EVALUATION ON THE BOTSWANA DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.6456	0.6641	0.6820	0.6564	0.2445	0.7091	0.5077	0.5225	0.7004	0.7706	0.7842	**0.8137**
P.	0.7158	0.7060	0.7417	0.6804	0.2876	0.7432	0.6022	0.5908	0.7485	0.8122	0.8002	**0.8442**
RS.	0.9361	0.9327	0.9300	0.9257	0.8289	0.9415	0.9042	0.9107	0.9420	0.9586	0.9515	**0.9607**
NMI	0.7416	0.7493	0.8324	0.6886	0.1805	0.7316	0.7050	0.6227	0.7676	0.8448	0.8573	**0.8624**
K.	0.6205	0.6416	0.6579	0.6331	0.1878	0.6875	0.4753	0.4867	0.6789	0.7529	0.7683	**0.8007**
V.	0.7416	0.7493	0.8324	0.6886	0.1805	0.7316	0.7050	0.6227	0.7676	0.8448	0.8573	**0.8624**
mean	0.7335	0.7405	0.7794	0.7121	0.3183	0.7574	0.6499	0.6260	0.7675	0.8306	0.8364	**0.8574**

TABLE VI
QUANTITATIVE EVALUATION ON THE PAVIAUNI DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.5033	0.5148	0.5872	0.5640	0.4272	0.6252	0.6584	0.3167	0.4981	0.4210	0.6907	**0.7161**
P.	0.6972	0.6988	0.7291	0.6631	0.4366	0.6726	0.7181	0.5457	0.6964	0.6667	**0.7355**	0.7301
RS.	0.7730	0.7784	0.7938	0.7565	0.2632	0.8322	0.8087	0.7360	0.7701	0.7627	0.8295	**0.8306**
NMI	0.5389	0.5633	0.6109	0.5501	0.0055	0.5079	0.5808	0.3040	0.5344	0.4923	0.6694	**0.6762**
K.	0.4720	0.4437	0.5064	0.4412	0.4259	0.5778	0.6368	0.2265	0.4204	0.3509	0.6620	**0.6806**
V.	0.5389	0.5633	0.6109	0.5501	0.0055	0.5079	0.5808	0.3040	0.5344	0.4923	0.6694	**0.6762**
mean	0.5872	0.5937	0.6397	0.5875	0.2607	0.6206	0.6639	0.4055	0.5756	0.5310	0.7094	**0.7183**

TABLE VII
QUANTITATIVE EVALUATION ON THE PAVIACENTRE DATASET

-	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
ACC.	0.6281	0.7614	0.8926	TO	TO	TO	0.5749	0.4331	0.4632	0.5712	0.9162	**0.9401**
P.	0.8836	0.8424	0.9196	TO	TO	TO	0.8167	0.7834	0.8576	0.9183	0.9312	**0.9502**
RS.	0.8460	0.9177	0.9675	TO	TO	TO	0.8052	0.7654	0.8120	0.8162	0.9862	**0.9880**
NMI	0.6908	0.7556	0.8931	TO	TO	TO	0.6113	0.4703	0.6708	0.7178	0.8946	**0.9323**
K.	0.5840	0.6980	0.8872	TO	TO	TO	0.5359	0.3815	0.3604	0.4922	0.9130	**0.9293**
V.	0.6908	0.7556	0.8931	TO	TO	TO	0.6113	0.4703	0.6708	0.7178	0.8946	**0.9323**
mean	0.7206	0.7884	0.9088	TO	TO	TO	0.6592	0.5507	0.6391	0.7056	0.9226	**0.9454**

TABLE VIII
DATASETS DESCRIPTION

Dataset	# Samples	# Clusters	# Dimension/Band
MNIST	70,000	10	784
Indian Pines	10,249	16	220
DFTC2013	2,832	15	144
Salinas	54,129	16	204
Botswana	3,248	14	145
Pavia Centre	148,152	9	102
Pavia University	42,776	9	103

negative pairs to be greater than the margin. SpecNet learns a map that embeds input data points into the eigenspace of their associated Laplacian matrix in a DL-based learning framework that involves constrained stochastic optimization.

For a fair comparison, the best results obtained by the processing components proposed in the above methods are presented. According to the results, we have the following observations.

1) DL-based clustering algorithms outperform nDL-based clustering algorithms in most cases. This is mainly due to the fact that deep learning technology provides a better learning ability for unsupervised learning. Most DL-based clustering algorithms optimize their models and perform the optimization in a stochastic fashion, allowing them to scale. The benefit of such an approach is available for large-scale datasets.
2) For nDL-based algorithms, the performance of KM and GMM is barely satisfactory as it is limited by the priori knowledge of data distribution. SC provides an unexpected result as it outperforms most nDL-based algorithms on real-world HSI datasets. SSC-based algorithms, including EnSC and SSC-BP, have a relative performance and show good potential for HSI datasets, but SSC-OMP does not provide a promising result. This may be because SSC-OMP is comparable with those derived for the original SSC only if the subspaces are sufficiently separated, and the data are well distributed.
3) For DL-based algorithms, AEC, VAE, and DEC are three AEC algorithms, and these three methods do not perform well in dealing with real-world HSI datasets. DEC is better than AEC and VAE since it iteratively optimizes a KL divergence-based clustering objective with a self-training target distribution. SiamNet, SpecNet, and our proposed DSCRLE outperform the other methods in most HSI datasets. The likely reason is that the above methods give more consideration to exploring data similarity. Our proposed DSCRLE shows better robustness and efficiency than SiamNet and SpecNet, as we introduce regularized linear embedding into DSCRLE, it contains both manifold structure information and linear-embedding constraints.
4) Fig. 2 illustrated the statistical evaluations of SpecNet, DSCRLE-woAGC, and DSCRLE, and each experiment is repeated ten times. DSCRLE-woAGC

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TABLE IX
RUNNING TIME ON TWO HSI DATASETS

	KM	GMM	SC	EnSC	SSC-OMP	SSC-BP	AEC	VAE	DEC	SiamNet	SpecNet	DSCRLE
Pavia Centre	5.75s	293.18s	2848.31s	TO	TO	TO	276.27s	483.99s	235.93s	569.04s	428.57s	563.64s
Salinas	0.96s	192.07s	444.94s	TO	TO	TO	173.28s	214.29s	117.62s	121.58s	249.91s	259.46s

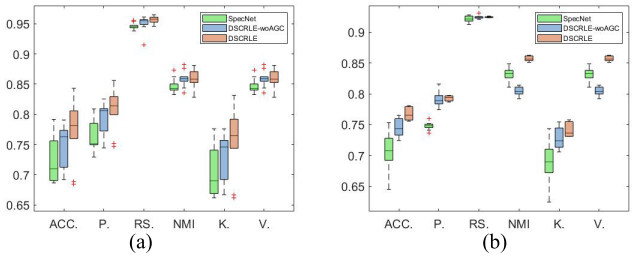

Fig. 2. Statistical evaluations of SpecNet, DSCRLE-woAGC, and DSCRLE. The loss function of DSCRLE-woAGC ignored regularized linear embedding, with only AGC. (a) Botswana. (b) Salinas.

and DSCRLE are better than SpecNet, since they improve the loss function by AGC, and DSCRLE outperforms DSCRLE-woAGC as regularized linear embedding provides a better representation for HSIs. Meanwhile, DSCRLE is more stable than SpecNet and DSCRLE-woAGC according to the box plot in Fig. 2.

E. Running Time

SC takes $O(N^2d)$ time for affinity matrix construction, where d and N are the number of feature dimensions and data points. Along with the increasing number N of data points, the time and space complexity of affinity matrix construction can be unsolvable problems. The affinity matrix $\mathbf{S}$ with AGC takes $O(Nkd)$ time for affinity matrix construction, and it is a significant improvement compared to graph-based clustering methods that need at least $O(N^2d)$. Regularized linear embedding takes $O(Nd^2)$ time for constructing a low-dimensional mapping function.

Focusing on the running time in Table IX, we compare the clustering algorithms on two large-scale datasets, including Pavia Centre and Salinas. KM is much more efficient since we use MiniBatchKMeans, which uses mini-batches to reduce the computation time. GMM is more efficient than SC, as SC-based algorithms struggle with the affinity matrix construction and the decomposition of Laplacian matrix for large-scale HSI datasets. SSC-based algorithms including EnSC, SSC-OMP, and SSC-BP are unavailable for large-scale HSI datasets as they take more than 1 h to data clustering. For DL-based algorithms, AEC and DEC outperform the other methods refer of time-consuming. SiamNet, SpecNet, and DSCRLE are similar, SiamNet needs to calculate the distance between positive and negative samples. SpecNet and DSCRLE take more time on affinity matrix construction as they need to calculate the distance of neighbors.

F. Effect of Spatial Information

Digital filtering is a data treatment method that enhances the signal-to-noise ratio of an analytical signal through the

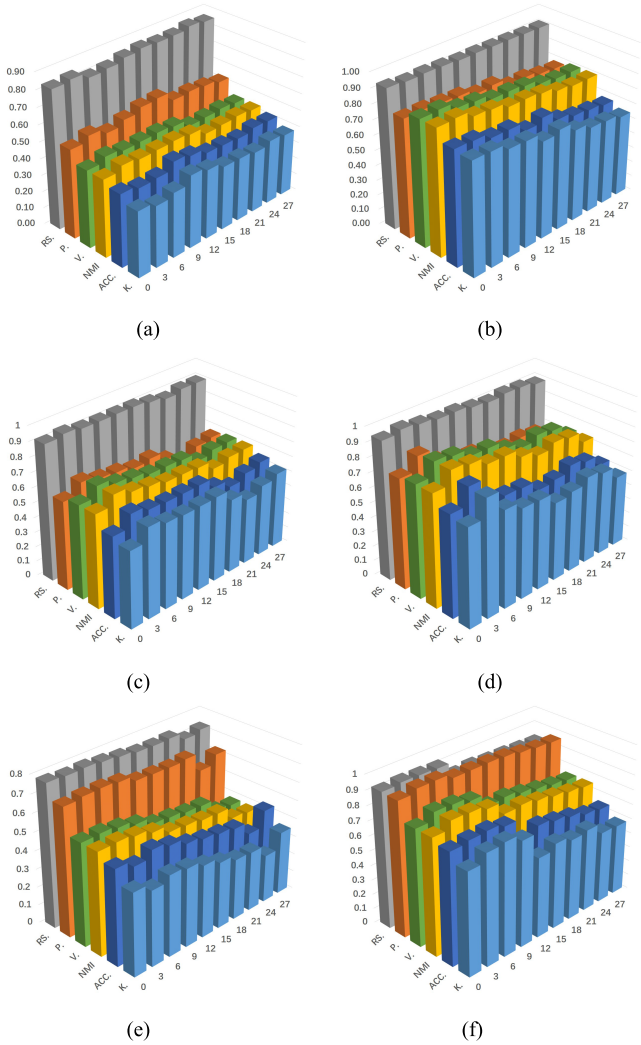

Fig. 3. Quantitative evaluation with different size of smoothing filters. (a) Indian Pines. (b) Salinas. (c) DFTC2013. (d) Botswana. (e) Pavia centre. (f) Pavia University.

convolution of a dataset with an appropriate filter. In this work, we introduce different smoothing filters to smooth the HSI images in the spatial domain and each pixel can be considered as the summary of its neighbors. As illustrated in Fig. 3, too small or too large size of smoothing filters do not provide good performance, and the size of the smoothing filter in the proposed DSCRLE is set to 5. The clustering maps and ground-truth maps of six HSI datasets are shown in Fig. 4, and it is evident that the above datasets contain many noise samples. One possible way to solve this problem is to introduce a robust dimensionality reduction or band selection to extract some representative features or bands, as it might be getting rid of the influences of noise samples. Otherwise, the proposed DSCRLE only

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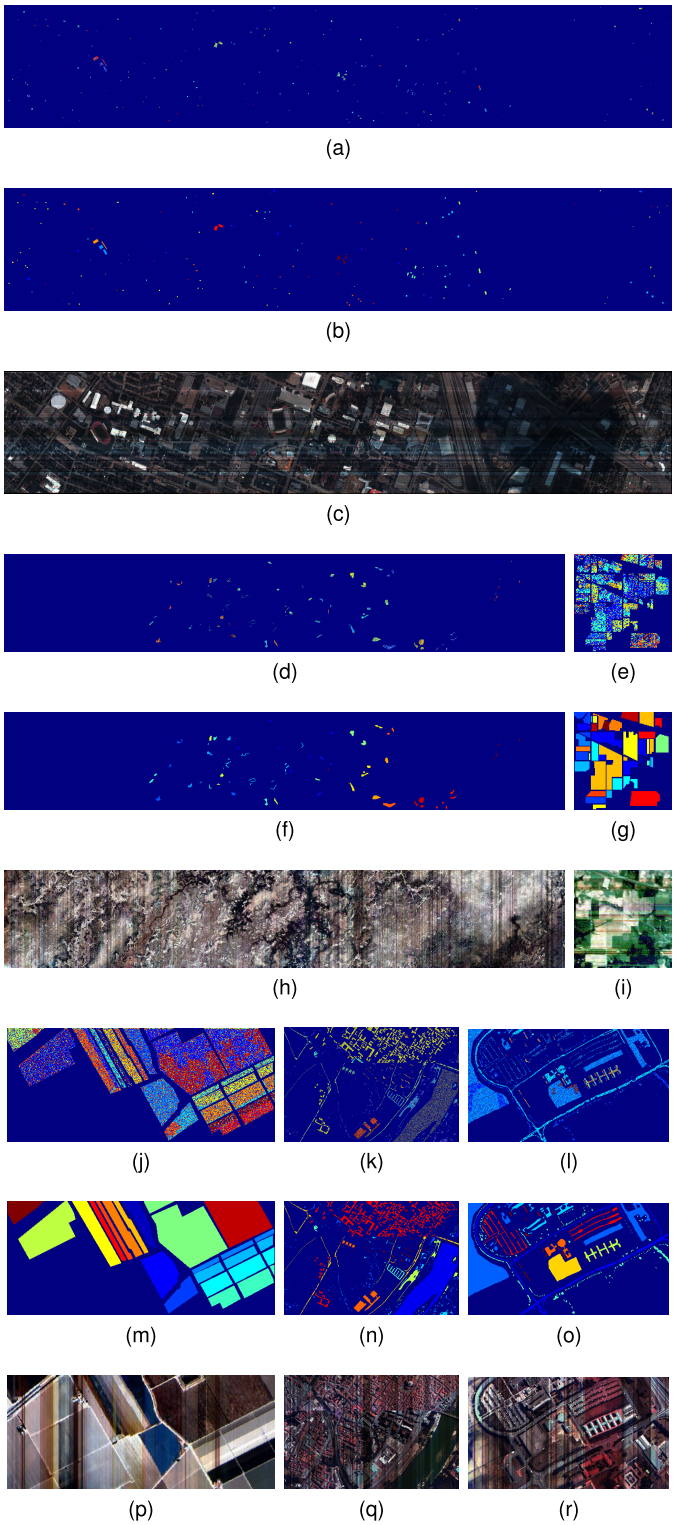

Fig. 4. Clustering maps, ground-truth maps, and visualizations of six HSIs. (a) Clustering maps of DFTC2013. (b) GT of DFTC2013. (c) Visualizations of DFTC2013. (d) Clustering maps of Botswana. (f) GT of Botswana. (h) Visualizations of Botswana. (e) Clustering maps of Indian Pines. (g) GT of Indian Pines. (i) Visualizations of Indian Pines. (j) Clustering maps of Salinas. (m) GT of Salinas, (p) Visualizations of Salinas. (k) Clustering maps of Pavia Centre. (n) GT of Pavia Centre. (q) Visualizations of Pavia Centre. (l) Clustering maps of Pavia University. (o) GT of Pavia University. (r) Visualizations of Pavia University.

adapts StandardScaler for data normalization, as the spatial information of HSIs is not fully utilized, and one possible method is to introduce some composition of different data argumentations.

V. Conclusion

In this work, we have presented a robust and efficient clustering approach named DSCRLE for large-scale HSI image processing. We implement a fully connected neural network with spectral orthogonality constraints, and it can perform the optimization in a stochastic fashion. It obtains a feasible solution for the eigenvalue decomposition of SC, which is available for scalability and generalization. Meanwhile, we explicitly introduce regularized linear embedding into SC for solving the problem of high-dimensional and sparse datasets. The proposed objective function contains both manifold structure information and linear embedding constraints. Extensive experiments have illustrated that the proposed DSCRLE outperforms well-known clustering algorithms on real-world HSI datasets.

Future work may include exploring some more efficient data argumentation, including positive-incentive noise and spectral–spatial information fusion.

1) Some cited references declared that the composition of multiple data argumentation can be beneficial to the clustering performance. Motivated by the existing work, we are considering the positive-incentive noise, which rethinks whether the noise always results in a negative impact [35].
2) Another problem is that the illustrated clustering maps have an obvious flaw similar to that of the phenomenon of salt and pepper. In our next work, we will learn to use spectral–spatial information fusion to improve the generalization capability of DSCRLE.

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